



4.2.4 Conditioning of the linear least squares problem



04 2 4 Conditioning of LLS

fit width

Given $A \in \mathbb{C}^{m \times n}$ with linearly independent columns and $b \in \mathbb{C}^m$, consider the linear least squares (LLS) problem

$$\|b - A\hat{x}\|_2 = \min_x \|b - Ax\|_2 \tag{4.2.1}$$

and the perturbed problem

$$\|(b + \delta b) - A(\hat{x} + \delta \hat{x})\|_2 = \min_x \|(b + \delta b) - A(x + \delta x)\|_2 \tag{4.2.2}$$

The question we want to examine is by how much the relative error in b is amplified into a relative error in $\hat{x}$. We will restrict our discussion to the case where A has linearly independent columns.

Now, we discovered that $\hat{b}$, the projection of b onto the column space of A , satisfies

$$\hat{b} = A\hat{x} \tag{4.2.3}$$

and the projection of $b + \delta b$ satisfies

$$\hat{b} + \delta \hat{b} = A(\hat{x} + \delta \hat{x}) \tag{4.2.4}$$

where $\delta \hat{b}$ equals the projection of δb onto the column space of A .

Let θ equal the angle between vectors b and its projection $\hat{b}$ (which equals the angle between b and the column space of A). Then

$$\cos(\theta) = \|\hat{b}\|_2 / \|b\|_2$$

and hence

$$\cos(\theta) \|b\|_2 = \|\hat{b}\|_2 = \|A\hat{x}\|_2 \leq \|A\|_2 \|\hat{x}\|_2 = \sigma_0 \|\hat{x}\|_2$$

which (as long as $\hat{x} \neq 0$) can be rewritten as

$$\frac{1}{\|\hat{x}\|_2} \leq \frac{\sigma_0}{\cos(\theta)} \frac{1}{\|b\|_2}. \tag{4.2.5}$$

Subtracting (4.2.3) from (4.2.4) yields

$$\delta \hat{b} = A\delta \hat{x}$$

or, equivalently,

$$A\delta \hat{x} = \delta \hat{b}$$

which is solved by

$$\begin{aligned} \delta \hat{x} &= A^\dagger \delta \hat{b} \\ &= A^\dagger A (A^H A)^{-1} A^H \delta b \\ &= (A^H A)^{-1} A^H A (A^H A)^{-1} A^H \delta b \\ &= A^\dagger \delta b, \end{aligned}$$

where $A^\dagger = (A^H A)^{-1} A^H$ is the pseudo inverse of A and we recall that $\delta \hat{b} = A(A^H A)^{-1} A^H \delta b$. Hence

$$\|\delta \hat{x}\|_2 \leq \|A^\dagger\|_2 \|\delta b\|_2. \tag{4.2.6}$$

Homework 4.2.4.1. Let $A \in \mathbb{C}^{m \times n}$ have linearly independent columns. Show that

$$\|(A^H A)^{-1} A^H\|_2 = 1/\sigma_{n-1},$$

where σ_{n-1} equals the smallest singular value of A .

[Hint](#) [Solution](#)

Let $A = U_L \Sigma_{TL} V^H$ be the reduced SVD of A , where V is square because A has linearly independent columns. Then

$$\begin{aligned}
& \| (A^H A)^{-1} A^H \|_2 \\
&= \\
& \| ((U_L \Sigma_{TL} V^H)^H U_L \Sigma_{TL} V^H)^{-1} (U_L \Sigma_{TL} V^H)^H \|_2 \\
&= \\
& \| (V \Sigma_{TL} U_L^H U_L \Sigma_{TL} V^H)^{-1} V \Sigma_{TL} U_L^H \|_2 \\
&= \\
& \| (V \Sigma_{TL}^{-1} \Sigma_{TL}^{-1} V^H) V \Sigma_{TL} U_L^H \|_2 \\
&= \\
& \| V \Sigma_{TL}^{-1} U_L^H \|_2 \\
&= \\
& \| \Sigma_{TL}^{-1} U_L^H \|_2 \\
&= \\
& 1/\sigma_{n-1}.
\end{aligned}$$

This last step needs some more explanation: Clearly

$\|\Sigma_{TL} U_L^H\|_2 \leq \|\Sigma_{TL}\|_2 \|U_L^H\|_2 = \sigma_0 \|U_L^H\|_2 \leq \sigma_0$. We need to show that there exists a vector x with $\|x\|_2 = 1$ such that $\|\Sigma_{TL} U_L^H x\|_2 = \|\Sigma_{TL} U_L^H\|_2$. If we pick $x = u_0$ (the first column of U_L), then $\|\Sigma_{TL} U_L^H x\|_2 = \|\Sigma_{TL} U_L^H u_0\|_2 = \|\Sigma_{TL} e_0\|_2 = \|\sigma_0 e_0\|_2 = \sigma_0$.

Use the reduced SVD of A .

Combining (4.2.5), (4.2.6), and the result in this last homework yields

$$\frac{\|\hat{\Delta x}\|_2}{\|\hat{x}\|_2} \leq \frac{1}{\cos(\theta)} \frac{\sigma_0}{\sigma_{n-1}} \frac{\|\mathcal{B}\|_2}{\|b\|_2}. \quad (4.2.7)$$

Notice the effect of the $\cos(\theta)b$. If b is almost perpendicular to $\mathcal{C}(A)$, then its projection $\hat{b}$ is small and $\cos \theta$ is small. Hence a small relative change in b can be greatly amplified. This makes sense: if b is almost perpendicular to $\mathcal{C}(A)$, then $\hat{x} \approx 0$, and any small $\Delta b \in \mathcal{C}(A)$ can yield a relatively large change Δx .

Definition 4.2.4.1. Condition number of matrix with linearly independent columns. Let $A \in \mathbb{C}^{m \times n}$ have linearly independent columns (and hence $n \leq m$). Then its condition number (with respect to the 2-norm) is defined by

$$\kappa_2(A) = \|A\|_2 \|A^\dagger\|_2 = \frac{\sigma_0}{\sigma_{n-1}}.$$

It is informative to explicitly expose $\cos(\theta) = \|\hat{b}\|_2 / \|b\|_2$ in (4.2.7):

$$\frac{\|\hat{\Delta x}\|_2}{\|\hat{x}\|_2} \leq \frac{\|b\|_2}{\|\hat{b}\|_2} \frac{\sigma_0}{\sigma_{n-1}} \frac{\|\mathcal{B}\|_2}{\|b\|_2}.$$

Notice that the ratio

$$\frac{\|\mathcal{B}\|_2}{\|b\|_2}$$

can be made smaller by adding a component, b_r , to b that is orthogonal to $\mathcal{C}(A)$ (and hence does not change the projection onto the column space, $\hat{b}$):

$$\frac{\|\mathcal{B}\|_2}{\|b + b_r\|_2}.$$

The factor $1/\cos(\theta)$ ensures that this does not magically reduce the relative error in $\hat{x}$:

$$\frac{\|\hat{\Delta x}\|_2}{\|\hat{x}\|_2} \leq \frac{\|b + b_r\|_2}{\|\hat{b}\|_2} \frac{\sigma_0}{\sigma_{n-1}} \frac{\|\mathcal{B}\|_2}{\|b + b_r\|_2}.$$